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Structured-sequential denoising for ring-based photoacoustic tomography

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Electromagnetic interference (EMI) in photoacoustic tomography (PAT) can arise from receive-chain electronics, grounding paths, laser/DAQ synchronization, and environmental coupling, and may appear as structured stripe artifacts and baseline drift in raw channel data. In this work, we propose a physics-motivated, structure-aware, learning-free Structured-Sequential Denoising (SSD) framework for ring-based PAT. SSD first suppresses quasi-synchronous cross-channel stripe-like interference using robust order-statistic processing and IQR-based time-index detection, and then performs single-channel temporal restoration to reduce baseline drift and residual random noise. Experiments on simulated and in vivo finger and zebrafish datasets show that SSD improves raw-signal and reconstructed-image quality while preserving acoustic signal fidelity under the tested ring-array setting. Compared with selected conventional, non-local, guided-filtering, and adaptive-filtering baselines, SSD achieved the best overall PSNR/SSIM on the simulation dataset and improved the mean A-line SNR by 19.21 dB and 24.05 dB over the noisy input across the finger and zebrafish datasets, respectively.

Electrical noise is a major source of signal corruption in photoacoustic tomography (PAT), comprising both residual thermal/electronic noise and parasitic electromagnetic interference (EMI). Thermal/electronic noise is commonly approximated as a zero-mean Gaussian component because the measured voltage fluctuation is the aggregate contribution of many microscopic random carrier motions in resistive and active receive-chain components. This Gaussian approximation is used only for the residual random electronic component, whereas structured EMI and baseline drift are treated as non-Gaussian components in this work. Parasitic EMI can arise from system electronics, grounding paths, acquisition circuitry, laser/DAQ synchronization, or environmental coupling, and may exhibit strong spatiotemporal coherence that cannot be effectively removed by standard frequency-domain denoising techniques[1,2].

Several denoising approaches for PAT have been shown to improve the signal-to-noise ratio (SNR) of PA signals[3–12]. Among these techniques, band-pass filtering[9] and signal averaging[3] are most commonly used; however, averaging requires multiple acquisitions, which is time-consuming and degrades temporal resolution. Moreover, because the spectra of EMI and true PA signals often overlap, band-pass filtering inevitably removes portions of the signal along with the noise, resulting in loss of structural detail. Researchers have also pursued denoising of raw A-lines using wavelet-based methods[4,6], empirical mode decomposition (EMD)[7,8], and sparsity-based approaches[13]. Nevertheless, these methods face one or more limitations: (i) the need of prior knowledge of noise characteristics, which is difficult to obtain in practice; and (ii) substantial computational cost, particularly for iterative optimization-based schemes.

In recent years, data-driven deep learning methods have introduced a paradigm shift in PAT image enhancement, achieving notable advances in denoising, artifact suppression, and image reconstruction[14,15]. Recent PA studies have explored a broad range of learning-based approaches, including U-Net-type networks, conditional generative adversarial networks, unsupervised and self-supervised denoising, neural-representation-based reconstruction, diffusion- or score-based generative priors, and physics-informed or physics-guided neural networks[14,16–23]. Nevertheless, these approaches also present practical limitations. Many supervised methods require large collections of paired, high-quality reference images, which are difficult or impossible to obtain for in vivo measurements[24]. Their performance may also degrade when the noise distribution encountered during

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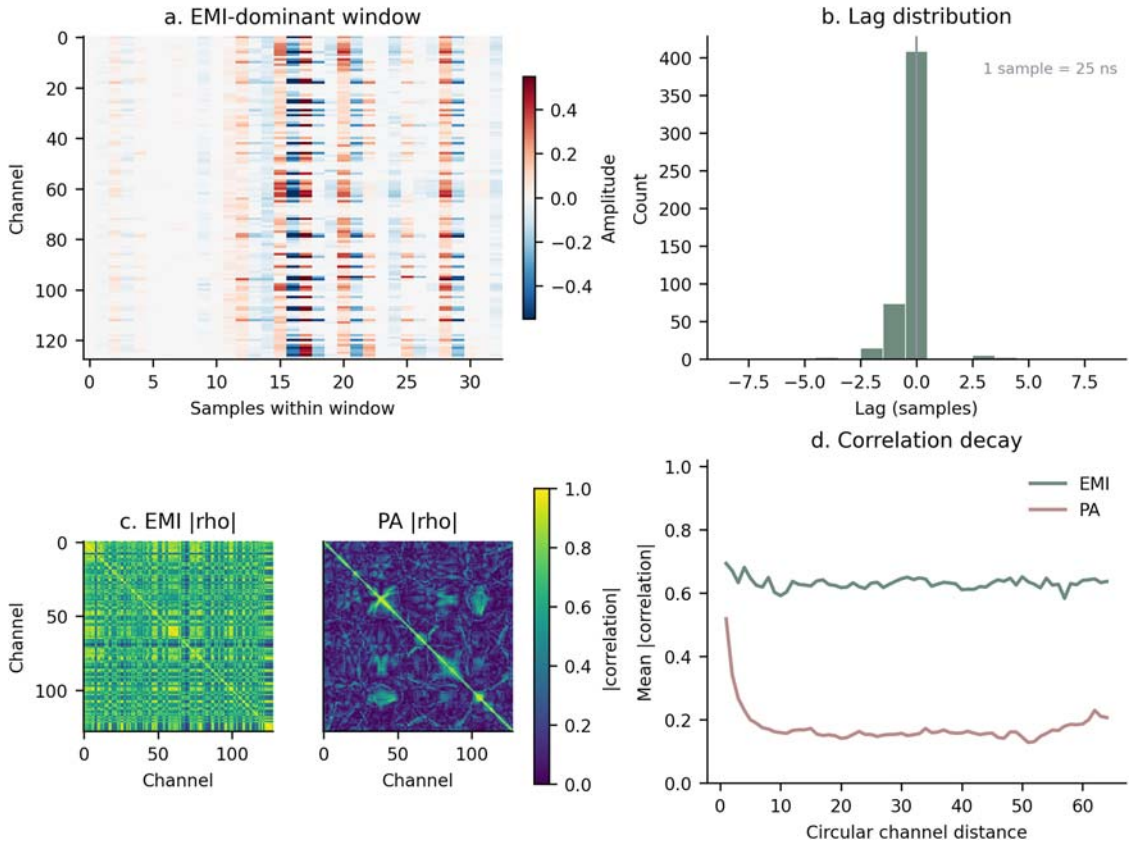


Fig. 1 | Quantitative validation of inter-channel EMI synchrony. a Representative EMI-dominant window extracted from the measured/background-derived stripe component. The window was centered at an IQR-selected event peak and fixed to 33 samples. **b** Histogram of normalized cross-correlation peak lags computed against a leave-one-out channel-averaged reference waveform. **c** Pairwise absolute inter-channel correlation heatmaps for EMI-dominant windows and clean PA-dominant control windows. **d** Mean absolute correlation as a function of circular channel distance.

inference differs from that represented in the training data, particularly for non-stationary or hardware-specific EMI[1]. Although unsupervised and self-supervised approaches reduce the dependence on paired clean data, they may still rely on assumptions regarding noise independence, repeated observations, physical-model accuracy, or consistency between the training and testing distributions[18,19,25]. Their robustness under structured and system-dependent EMI therefore requires further investigation.

To further improve EMI suppression in raw PAT data, we propose a physics-motivated, structure-aware, learning-free framework, termed Structured-Sequential Denoising (SSD). SSD is not intended as a new globally optimal denoising theory. Instead, it organizes standard robust filtering, IQR-based outlier detection, baseline correction, and adaptive Kalman filtering according to the measured structure of raw multi-channel PAT data. The first stage targets quasi-synchronous cross-channel stripe-like interference, while the second stage performs single-channel temporal restoration to reduce baseline drift and residual random electronic noise. The contribution of this work lies in the PAT-specific sequencing and integration of these operations, together with quantitative validation of EMI synchrony, false-positive behavior, signal preservation, and pre-beamforming fidelity.

Here, we present SSD, a structure-aware and learning-free signal-domain framework for EMI suppression in ring-based PAT. SSD exploits the quasi-synchronous cross-channel structure of stripe-like EMI before applying single-channel temporal restoration for baseline drift and residual random noise. We validate the framework on simulated vascular data, in vivo human finger data, and live zebrafish data, and further evaluate EMI synchrony and false-positive behavior.

Results

Quantitative validation of inter-channel EMI synchrony

To examine the inter-channel structure used by SSD, we analyzed the synchrony of the experimentally measured background EMI component. EMI-dominant events were identified from the column-wise RMS profile using an IQR-based threshold. For each detected event window, the normalized cross-correlation peak lag of each channel was calculated relative to a leave-one-out channel-averaged reference waveform (Fig. 1a, b).

The lag distribution was concentrated near zero on the sampling grid. The median, IQR, and standard deviation were 0, 0, and 0.83 samples, respectively. Overall, 79.7% of the channel-event pairs had a zero-sample peak lag, and the maximum absolute lag was 6 samples (Fig. 1b). This indicates that the dominant EMI events were largely synchronized across channels and did not follow the systematic channel-dependent delay pattern expected from acoustic propagation.

We also compared the pairwise inter-channel correlations of EMI-dominant windows and PA-dominant control windows. The mean off-diagonal absolute correlation was higher for EMI than for PA signals (0.63 ± 0.18) versus (0.17 ± 0.15). The EMI correlation also changed little with circular channel distance, decreasing from 0.69 at ($d = 1$) to 0.6365 at ($d = 64$). In contrast, the PA correlation decreased from 0.52 to 0.21 over the same distance range (Fig. 1c, d). These results support the use of cross-channel structure, rather than signal amplitude alone, as the main criterion for identifying stripe-like EMI in the present dataset.

To avoid overextending the main text with validation details, additional information on this synchrony analysis is provided in the Supplementary Information.

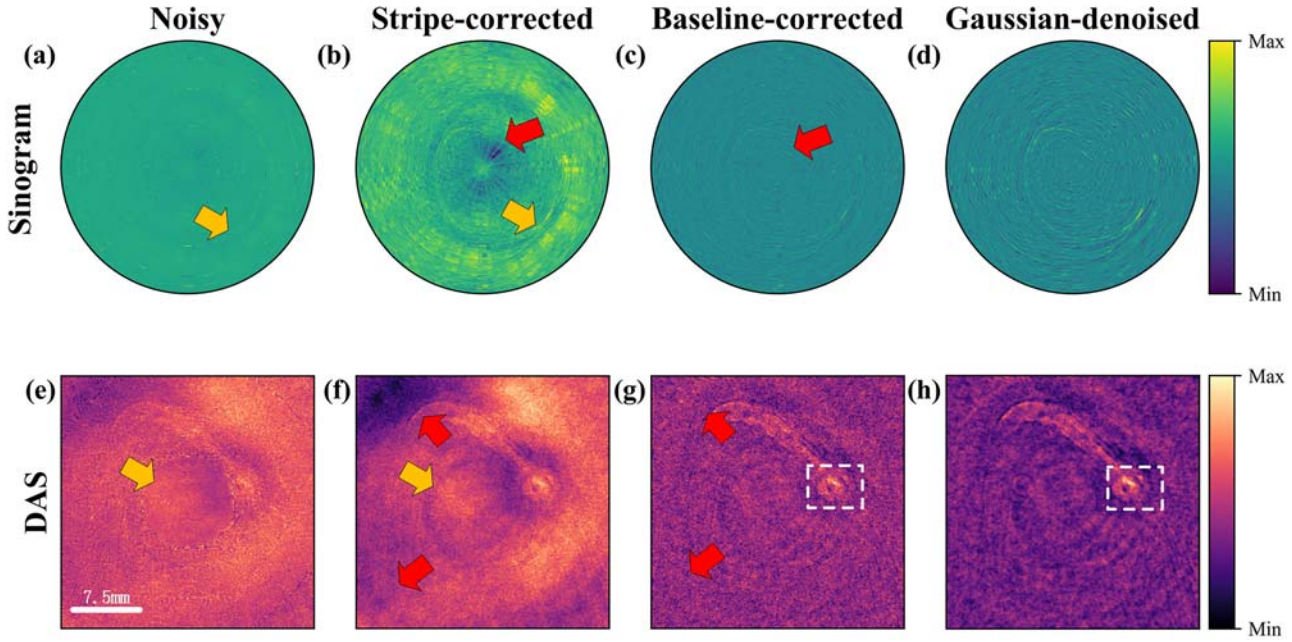


Fig. 2 | Stepwise denoising effects on reconstructed images and their corresponding sinograms. a–d Sinograms showing progressive attenuation of low-frequency trends, reduction of stripe energy, and suppression of residual high-frequency noise. **e–h** Raw reconstructions corresponding to (**a–d**), demonstrating improved image contrast and structural clarity.

Step-wise noise suppression

To establish the method's effectiveness and robustness under realistic acquisition conditions, we begin with stepwise validation on experimental data, attributing the cumulative gains of each suppression step. We prioritize experimental data because it naturally embodies real-world system nonidealities-including inter-channel inconsistencies, baseline drift, and mixed noise-thereby demonstrating practical relevance and generalizability. We then complement these results with simulations, where ground truth is available, to further quantify the improvement contributed by each step.

As indicated by the yellow arrows in Fig. 2a, strong vertical stripe artifacts dominate the raw sinogram, significantly reducing local contrast and obscuring PA echoes. Correspondingly, in the DAS-reconstructed zebrafish image obtained from the raw data (Fig. 2e, yellow arrows), the body contour is barely discernible and accompanied by distinct ring-like artifacts. After stripe correction, which specifically targets structured stripes aligned with the channel direction, the stripe energy is substantially attenuated, and the contrast improves markedly. In Fig. 2b, the stripes at the yellow-arrow positions are notably weakened, whereas the red arrows still highlight residual low-frequency baseline drift and approximately Gaussian random noise. As shown in Fig. 2e, f, the outer boundary of the zebrafish becomes much clearer, and the ring artifact indicated by the yellow arrows in Fig. 2e is effectively removed. However, background nonuniformity caused by drift and random noise continues to limit the overall image contrast.

As shown in Fig. 2b, the red arrows indicate evident baseline drift. The baseline-drift correction step estimates inter-channel baseline offsets to restore cross-channel consistency. After correction (Fig. 2c), the baseline drift at the red-arrow positions is greatly reduced. As shown in Fig. 2f, g, background nonuniformity is mitigated, the dark background pointed out by the red arrows is effectively removed, and background consistency is improved. The continuity and visibility of the zebrafish structure are enhanced, and overall image contrast increases. Nevertheless, residual random noise still softens the edges of fine vessels and the bilateral eye regions, as highlighted by the white dashed box in Fig. 2g.

Finally, as shown in Fig. 2c, d, the final recursive filtering step suppresses the residual high-frequency and Gaussian noise, thereby improving the overall SNR of the sinogram while effectively preserving genuine structural textures and edge details. In the reconstructed images (Fig. 2g, h), edges become sharper and fine structures (such as microvessels) are more clearly distinguishable. As highlighted by the white dashed boxes in Fig. 2g, h, the local contrast is markedly enhanced.

Simulated vascular dataset

We then compared the results of the proposed SSD method with its counterparts in a representative simulated vascular scan for a systematic performance assessment (Fig. 3).

Quantitatively, SSD achieved the best performance in both the sinogram and DAS reconstruction domains. Compared with the noisy input, SSD improved PSNR from 13.21 dB to 25.91 dB in the DAS domain and from 16.97 dB to 36.32 dB in the sinogram domain. SSIM increased from 0.253 to 0.862 in the DAS domain and from 0.703 to 0.944 in the sinogram domain.

As shown in Fig. 3, the noisy input contains low-frequency baseline drift, structured stripe noise, and high-frequency random fluctuations. 1D-GF reduces part of the stripe-like components but leaves visible residual artifacts. BM3D mainly suppresses local random fluctuations and has limited effect on globally correlated stripes. WLSF reduces some background variations but over-smooths vessel-like structures. NLM provides generic patch-wise smoothing but does not explicitly target quasi-synchronous EMI. NLMS-ANC shows limited improvement because no independent reference noise channel is available. In contrast, SSD suppresses stripe artifacts and baseline drift while preserving the continuity of PA echo trajectories.

Quantitatively, SSD achieved the best performance in both the sinogram and DAS reconstruction domains. Compared with the noisy input, SSD improved PSNR from 13.21 dB to 25.91 dB in the DAS domain and from 16.97 dB to 36.32 dB in the sinogram domain. SSIM increased from 0.253 to 0.862 in the DAS domain and from 0.703 to 0.944 in the sinogram domain.

Taken together, the simulation results indicate that SSD enables component-wise suppression of EMI constituents (stripes + baseline) and high-frequency thermal noise in the signal domain, while retaining the meaningful PA content.

To further visualize the effect of each method on EMI-related components, we analyzed the denoised sinograms in the 2–D frequency domain.

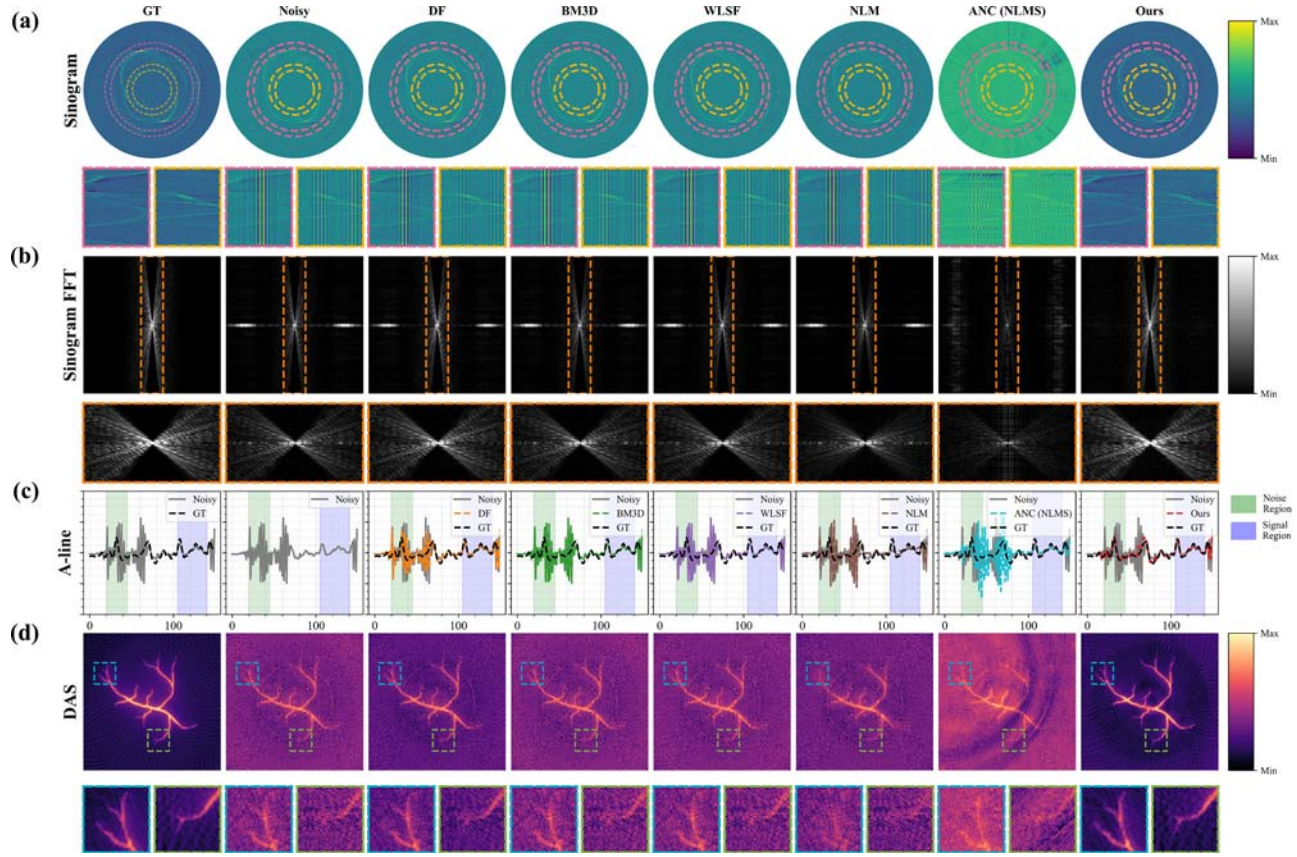


Fig. 3 | Visual comparison of different EMI denoising methods on simulated PA data. Columns from left to right show GT, noisy input, DF (1D-GF), BM3D, WLSF, NLM, ANC (NLMS; NLMS-ANC), and Ours (SSD). **a** Sinogram images, with two zoom-in insets cropped from the red and golden ROIs; arrows mark EMI artifacts targeted for removal. **b** Two-dimensional Fourier spectra of the sinograms in (**a**), with the central frequency band enlarged to improve visibility. **c** Representative A-line waveforms highlighting the PA signal versus background before and after denoising. The A-line is taken from the detector channel at 90° (ID #64/128), selected for its representative SNR and EMI pattern. **d** Corresponding DAS reconstructions, with two zoom-in insets highlighted in the tested baselines by green and cyan boxes.

Table 1 | Quantitative comparison of denoising methods on the current simulated GT-plus-noise data

Metric	Noisy	1D-GF	BM3D	WLSF	NLM	NLMS-ANC	SSD
PSNR (DAS) [dB]	13.21	16.71	13.20	13.36	14.06	10.55	**25.91**
PSNR (Sinogram) [dB]	16.97	17.78	16.96	16.97	17.90	10.10	**36.32**
SSIM (DAS) [–]	0.253	0.388	0.253	0.256	0.195	0.221	**0.862**
SSIM (Sinogram) [–]	0.703	0.762	0.703	0.719	0.654	0.540	**0.944**

As shown in Fig. 3b, the noisy sinogram exhibits stripe-related horizontal spectral components and background spectral clutter outside the main vessel-related support. 1D-GF and WLSF attenuate part of these stripe-related components but leave visible residual ridges. BM3D and NLM mainly reduce diffuse local fluctuations and have limited effect on globally correlated stripe artifacts. NLMS-ANC shows limited suppression in this setting because no independent hardware reference noise channel is available. In contrast, SSD more effectively suppresses stripe-related spectral components while preserving the main PA–related spectral support, which is consistent with its superior quantitative performance in Table 1.

Signal-domain differences directly propagate to the reconstructed image domain. Residual stripes, baseline drift, and high-frequency fluctuations in the sinogram can manifest after DAS reconstruction as ring-like artifacts, background non-uniformity, and speckle-like noise. Compared with the noisy input, SSD improved PSNR from 13.21 dB to 25.91 dB in the DAS domain and from 16.97 dB to 36.32 dB in the sinogram domain. SSIM increased from 0.253 to 0.862 in the DAS domain and from 0.703 to 0.944 in the sinogram domain. These results indicate that SSD suppresses structured EMI while preserving useful PA signal content better than the tested baselines.

To quantify potential over-suppression by the stripe-correction stage, we evaluated the false-positive rate and signal/EMI energy changes on the simulation dataset, where both the clean PA signal and injected EMI component were known. At the adopted operating point (α = 2.5), the method achieved an FPR of 1.99%, a PA signal energy loss of 2.39%, and an EMI removal ratio of 97.11%, indicating a favorable trade-off between EMI suppression and PA signal preservation.

As further support for this trade-off, threshold-sensitivity, signal-preservation and raw-domain fidelity analyses are provided in the Supplementary Information.

In vivo finger dataset

We then applied the method to a more challenging in vivo finger PA dataset to assess its robustness under real-world conditions. Fig. 4a illustrates the method-independent ROI annotation used for gCNR calculation. The signal and background ROIs were independently defined by a blinded

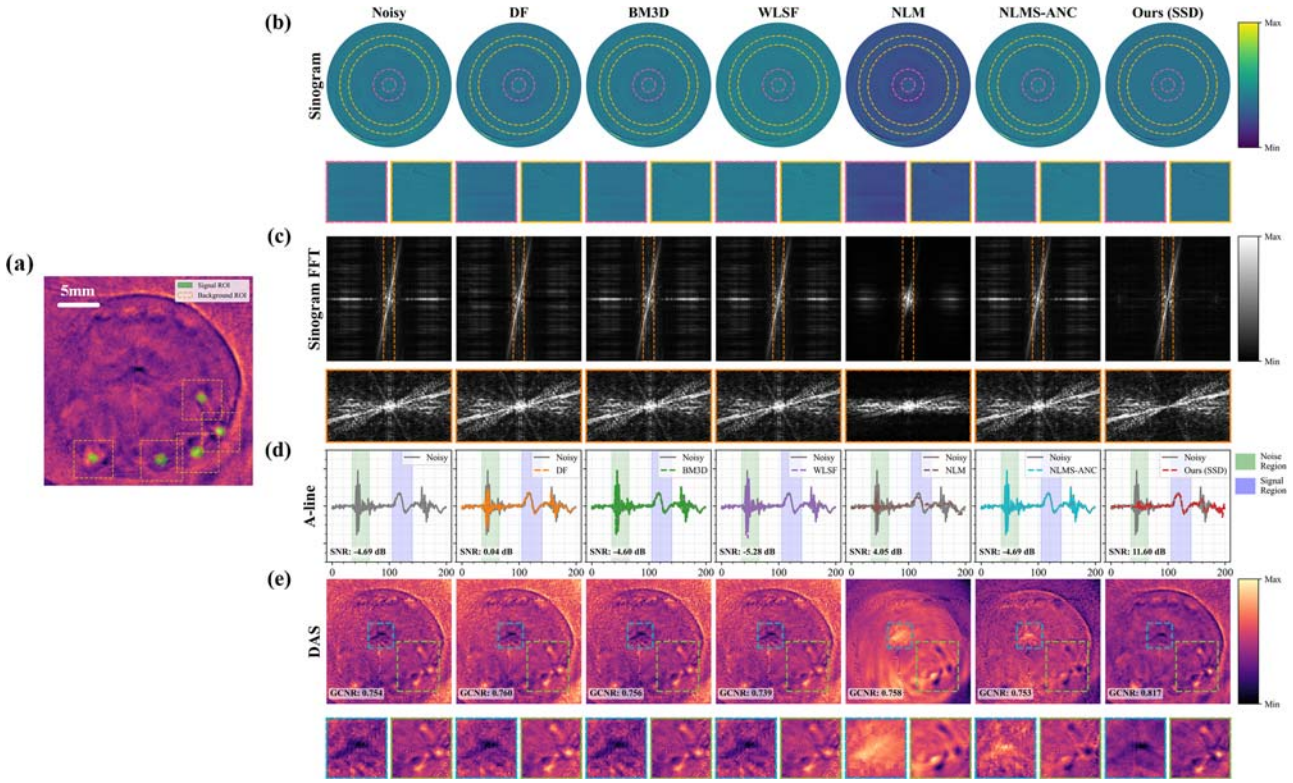


Fig. 4 | Performance of different EMI denoising methods on finger PA data. **a** Reference mask used for gCNR calculation. The green-shaded region indicates the signal ROI, and the golden dashed box denotes the background ROI. **b** Denoised sinograms from left to right: noisy input, DF (1D-GF), BM3D, WLSF, NLM, NLMS-ANC, and Ours (SSD). Two zoom-in insets are cropped from the red and golden ROIs; arrows mark EMI artifacts targeted for removal, including stripe patterns and baseline drift. **c** Two-dimensional Fourier spectra of the sinograms in (**b**). The central frequency band is cropped and enlarged to better visualise EMI-related stripe components in the angular--frequency domain. **d** Representative A-line waveforms highlighting the PA signal versus background noise. Each panel is annotated with its SNR. The A-line is taken from the detector channel at 90° (ID #64/128), selected for its representative SNR and EMI pattern. **e** Corresponding DAS-reconstructed images for (**b**). Zoom-in insets highlight the same ROIs marked by green and cyan boxes. The gCNR value is annotated at the lower-left corner of each image.

evaluator using only the raw/noisy DAS image, and the same fixed masks were applied unchanged to all methods.

In the signal domain (Fig. 4b), the noise patterns are more complex. SSD remains robust, yielding a cleaner sinogram background and better delineation of target structures. From the magnified regions in Fig. 4b and the representative A-lines in Fig. 4d, SSD effectively suppresses structured stripe artifacts (indicated by red arrows) and markedly removes the black horizontal bands associated with baseline drift (see the insets of Fig. 4b), while preserving continuous PA-echo trajectories without over-smoothing. In contrast, 1D–GF reduces stripe energy but leaves residual stripes as well as baseline drift and thermal noise, while BM3D and WLSF perform poorly on this dataset, offering little effective denoising effect.

The corresponding 2-D FFT magnitude maps of the sinograms (Fig. 4c) provide a complementary frequency-domain view. In the noisy data, EMI stripes give rise to strong horizontal spectral bands superimposed on the X-shaped support produced by the vessel structures. 1D–GF and WLSF attenuate these bands but leave clear residual ridges, whereas BM3D mainly suppresses diffuse high-frequency components. In contrast, SSD substantially reduces the horizontal spectral bands while keeping the main X-shaped passband, which is consistent with its cleaner sinogram appearance.

Across 17 finger frames, SSD achieved the highest mean performance among the tested methods, with a gCNR of 0.801 ± 0.053 and an A-line SNR of 17.477 ± 5.052 dB. Compared with the strongest non-SSD baselines, SSD improved gCNR from 0.771 ± 0.048 for 1D–GF to 0.801 ± 0.053, and improved A-line SNR from 7.995 ± 8.556 dB for NLM to 17.477 ± 5.052 dB. These frame-wise results indicate that the improvement observed in the representative finger image is consistent across repeated frames.

In vivo zebrafish dataset

The zebrafish dataset is characterized by stronger, non-stationary EMI and heavier thermal noise.

As shown in Fig. 5b, c, the outcomes are consistent with the preceding experiments. We annotate representative noise and signal locations in Fig. 5b (red arrows denote PA echoes). Even under high-noise conditions, the proposed SSD yields the highest-quality sinograms, markedly suppressing all three EMI constituents (stripe artifacts, baseline drift, and high-frequency components).

The corresponding 2–D FFT magnitude maps of the sinograms (Fig. 5c) provide a complementary frequency-domain view. The noisy data exhibit dense horizontal spectral bands and strong spectral clutter around the origin, reflecting the severe EMI and thermal noise in this dataset. 1D–GF, BM3D, and WLSF slightly reduce the diffuse background energy, but the horizontal bands and low-frequency ridges remain conspicuous. In contrast, SSD substantially attenuates the horizontal spectral bands while preserving the central structural support, which is consistent with its visually cleaner sinograms.

Overall, 1D-GF, BM3D, and WLSF provide only limited suppression of stripe artifacts and baseline drift. The A-line SNR trends align with these observations (Fig. 5d). The reconstructed images (Fig. 5e) exhibit a flatter background and substantially fewer ring artifacts, with cranial details becoming more discernible. Across 18 zebrafish frames, SSD remained higher than the strongest non-SSD baseline, achieving a gCNR of 0.454 ± 0.050 and an A-line SNR of 13.417 ± 2.684 dB. The strongest non-SSD baseline was NLM for both metrics, with a gCNR of 0.273 ± 0.060 and an A-line SNR of 0.995 ± 5.582 dB. These frame-wise results support the robustness of SSD under the stronger and more non-stationary EMI

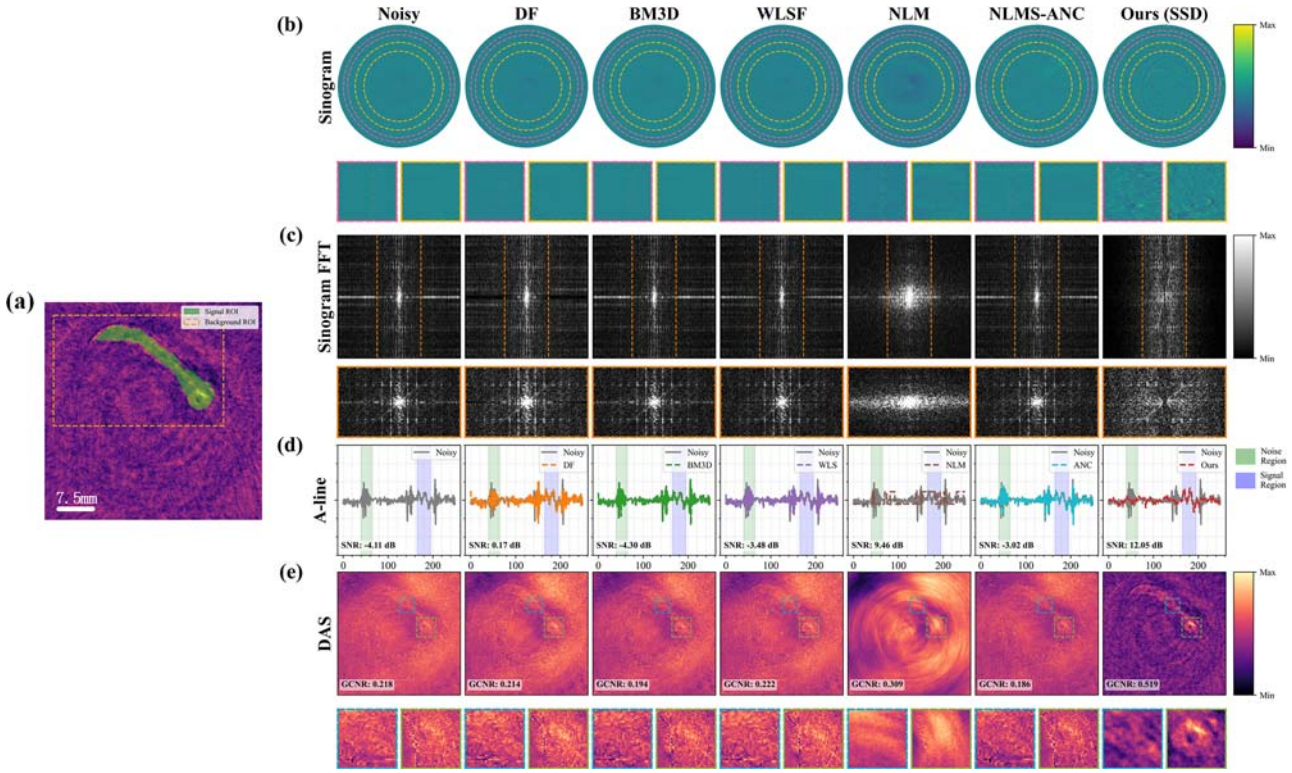


Fig. 5 | Performance of different EMI denoising methods on zebrafish PA data. **a** Reference mask used for gCNR calculation. The green-shaded region indicates the signal ROI, and the golden dashed box denotes the background ROI. **b** Denoised sinograms from left to right: noisy input, DF (1D-GF), BM3D, WLSF, NLM, NLMS-ANC, and Ours (SSD). Two zoom-in insets are cropped from the red and cyan ROIs; arrows mark EMI artifacts targeted for removal, including stripe patterns and baseline drift. **c** Two-dimensional Fourier spectra of the sinograms in (**b**). The central frequency band is cropped and enlarged to better visualise EMI-related stripe components in the angular--frequency domain. **d** A-line waveforms highlighting the PA signal versus background noise. Each panel is annotated with its SNR. The A-line is taken from the detector channel at 90° (channel 64), selected for its representative SNR and EMI pattern. **e** Corresponding DAS-reconstructed images for (**b**). Zoom-in insets highlight the same ROIs marked by red and cyan boxes. The gCNR value is annotated at the lower-left corner of each image.

condition in the zebrafish experiment. The gCNR and A-line SNR values for both in vivo datasets are summarized in Table 2.

Discussion

This study presents SSD, a structure-aware and learning-free signal-domain framework for suppressing EMI in ring-based PAT. The proposed pipeline targets three raw-data degradation components: quasi-synchronous stripe-like EMI, baseline drift, and residual random electronic noise. By estimating interference-related statistics directly from the measured raw data rather than relying on paired clean training labels, SSD is suitable for the tested low-SNR in vivo datasets and can be used as a practical preprocessing strategy.

The main advantage of SSD is that it operates before image reconstruction. EMI-related artifacts are identified and corrected in the raw channel–time domain, where quasi-synchronous cross-channel interference can be distinguished from channel-dependent acoustic PA propagation. This signal-domain design reduces the risk of correcting reconstruction-induced artifacts rather than the underlying measurement corruption. In the human finger and zebrafish experiments, both acquired using a 128-channel circular transducer under noticeable EMI contamination, SSD improved raw-signal and reconstructed-image quality without modifying the acquisition protocol.

Quantitative and qualitative evaluations on simulated, human finger, and zebrafish datasets demonstrated consistent improvements under the tested ring-array setting. SSD achieved the best overall PSNR/SSIM performance on the simulation dataset and improved A-line SNR on the in vivo datasets. The simulation-based false-positive, spectral-fidelity, and pre-beamforming analyses further indicated a favorable trade-off between EMI suppression and PA signal preservation. Together, these results support SSD as an effective signal-domain preprocessing framework for the tested ring-based PAT setting.

The performance of SSD is primarily determined by the extent to which the dominant interference exhibits a cross-channel structure distinguishable from acoustic PA propagation. In systems where the interference becomes highly asynchronous, spatially localized, or characterized by channel-dependent delays, the discriminative power of the current cross-channel statistics may be reduced. Similarly, when weak PA signals and structured EMI exhibit substantial overlap in both temporal and channel-domain characteristics, more conservative parameter settings or additional system-specific information may be required to maintain reliable separation. The current temporal-restoration stage is optimized for smoothly varying baseline dynamics, and its performance under abrupt baseline transitions or rapidly varying interference could be further improved using more flexible state models. Therefore, transfer of SSD to a new acquisition platform should include verification of the interference structure and system-specific adjustment of the parameters.

SSD is also computationally lightweight. For a typical 128-channel dataset, the denoising/preprocessing step can be completed within a short runtime on a standard workstation without dedicated parallelization, excluding DAS reconstruction unless otherwise stated. This makes the method compatible with practical preprocessing workflows and suggests potential integration with model-based or learning-based reconstruction pipelines. By reducing structured corruption at the signal level, SSD may improve the conditioning of subsequent reconstruction and reduce the burden on downstream image-domain correction.

Table 2 | Frame-wise quantitative comparison of gCNR and A-line SNR achieved by different denoising methods on the in vivo finger and zebrafish datasets

Dataset	Method	gCNR [–]	SNR (A-line) [dB]	*n*
Finger	Noisy	0.751 ± 0.054	−1.730 ± 3.432	17
	1D-GF	0.771 ± 0.048	4.361 ± 5.103	17
	BM3D	0.752 ± 0.055	−1.730 ± 3.432	17
	WLSF	0.745 ± 0.055	−1.540 ± 3.571	17
	NLM	0.732 ± 0.055	7.995 ± 8.556	17
	NLMS-ANC	0.751 ± 0.055	−1.730 ± 3.432	17
	SSD	0.801 ± 0.053	17.477 ± 5.052	17
Zebrafish	Noisy	0.245 ± 0.059	−10.634 ± 7.548	18
	1D-GF	0.262 ± 0.059	−5.323 ± 7.084	18
	BM3D	0.245 ± 0.059	−10.639 ± 7.549	18
	WLSF	0.251 ± 0.060	−10.270 ± 8.446	18
	NLM	0.273 ± 0.060	0.995 ± 5.582	18
	NLMS-ANC	0.244 ± 0.047	−9.826 ± 7.977	18
	SSD	0.454 ± 0.050	13.417 ± 2.684	18

Values are reported as mean ± standard deviation across repeated frames. Image-domain ROIs were independently defined by a blinded evaluator using the raw/noisy DAS images and then applied unchanged to all methods for gCNR computation. A-line SNR was computed using the same selected one-dimensional signal and background/noise windows for all methods within each frame.

The corresponding per-frame runtime measurements are provided as supplementary results in the Supplementary Information.

Future work will focus on extending SSD to broader acquisition settings and interference regimes. Important directions include validation on other array geometries and receive-chain designs, adaptive thresholding guided by system-specific noise statistics, nonlinear state estimation for abrupt baseline variations, and hybrid integration with learning-based reconstruction or denoising models. These extensions may improve robustness while preserving the main advantage of SSD: interpretable, training-free EMI suppression directly in the measured signal domain.

Methods

Signal delay characteristics

In PAT, when a biological tissue is irradiated by a pulsed laser excitation, the generated PA waves propagate from the optical absorber to the ultrasonic (US) transducer array[26]. During propagation, the PA waves first traverse heterogeneous biological tissue, and then pass through a coupling medium with more uniform acoustic properties (e.g., water). This propagation exhibits strong spatially dependent delay characteristics due to the continuously varying distances between the source and each transducer element. That is, the arrival times of the PA signals are progressively delayed across the transducer elements of the array[27].

In contrast, the dominant EMI observed in our system appears as a near-synchronous and strongly cross-channel-correlated transient in the raw channel–time data. We do not assume that this interference originates exclusively from transducer-element pickup. Instead, it may be coupled through shared receive-chain electronics, grounding paths, cables, acquisition circuitry, laser/DAQ synchronization, or environmental pathways. Once recorded, such interference can corrupt the subsequent beamforming process regardless of its exact hardware coupling point.

The distinction exploited in this work is therefore not the absolute temporal bandwidth of the waveform, nor a purely theoretical sub-nanosecond propagation argument. Rather, it is the measured inter-channel structure: true PA signals are governed by acoustic propagation and exhibit detector-dependent time-of-arrival patterns, whereas the dominant stripe-like EMI component behaves as a quasi-common-mode event across many channels. This structural difference motivates the first-stage cross-channel stripe-detection and correction module.

To visualize the different raw-data manifestations of PA signals and stripe-like EMI, representative channel–time measurements are shown in Fig. 6b, c. EMI can occur at arbitrary times within the acquisition window and therefore cannot be removed by simple temporal cropping. The key distinction used by SSD is the measured cross-channel structure: stripe-like EMI appears as a near-synchronous event across many channels, whereas PA signals follow detector-dependent acoustic propagation patterns.

Raw-data signal model and structural assumption

Let $X(c, t)$ denote the measured raw channel data, where $c = 1, \ldots, N$ indexes the receive channel and $t = 1, \ldots, T$ indexes the temporal sample. We model the measured signal as

$$X_c(t) = P_c(t) + E_c(t) + B_c(t) + N_c(t), \tag{1}$$

where $P_c(t)$ denotes the acoustic PA signal, $E_c(t)$ denotes the EMI-related transient component, $B_c(t)$ denotes slowly varying baseline drift, and $N_c(t)$ denotes residual random electronic noise.

For the dominant stripe-like EMI component observed in this study, we use a quasi-common-mode receive-chain model:

$$E_c(t) = a_c e(t - \delta_c), \tag{2}$$

where $e(t)$ is a shared interference waveform, a_c represents channel-dependent coupling strength, and δ_c represents small inter-channel timing jitter. This model does not specify a unique hardware coupling point. The interference may be coupled through shared electronics, grounding paths, cable shielding, acquisition circuitry, or other receive-chain components. The proposed method targets the data-domain manifestation of this interference, namely its near-synchronous cross-channel structure.

The residual random electronic component $N_c(t)$ is approximated as Gaussian only after structured EMI and deterministic baseline components are separated. Therefore, this work does not assume that the entire raw noise process is Gaussian. Thus, the conventional frequency-domain filters are ineffective for EMI suppression.

To address the coupling challenge introduced in the previous section, it is essential to distinguish between EMI and the true PA signal. A key characteristic exploited here is that EMI tends to appear simultaneously across all the channels of the receiving array, whereas the genuine PA signal exhibits slight inter-channel time delays due to differences in propagation paths. By leveraging this temporal disparity, it becomes feasible to effectively separate EMI from the acquired PA signals.

To clarify this principle, we first provide a detailed theoretical foundation for the proposed de-noising method. In the forward model for PAT in a homogeneous medium, the signal detected by the transducer at the location $\mathbf{r}_n$ and time t is expressed as[28,29]

$$\hat{p}(\mathbf{r}_n, t) = \frac{1}{4\pi c^2} \sum_{m=1}^{M} v_m \frac{p_0(\mathbf{r}'_m)}{\|\mathbf{r}'_m - \mathbf{r}_n\|} h'_e\left(t - \frac{\|\mathbf{r}'_m - \mathbf{r}_n\|}{c}\right) \quad n = 1, 2, \ldots, N, t \geq 0 \tag{3}$$

Here, we have M point sources distributed at $\mathbf{r}'_m$, $m = 1, 2, \ldots, M$, and N transducer elements distributed at $\mathbf{r}_n$, $n = 1, 2, \ldots, N$; c is the speed of sound; v_m is the volume of the m-th source point; $p_0(\mathbf{r}'_m)$ is the initial pressure at $\mathbf{r}'_m$; $h_e(t)$ is the ultrasonic transducer's electric impulse response (EIR), and $h'_e(t)$ denotes its time derivative.

For PA signals, the channel-dependent delay can be seen directly from the forward model. In a homogeneous medium, the signal received at

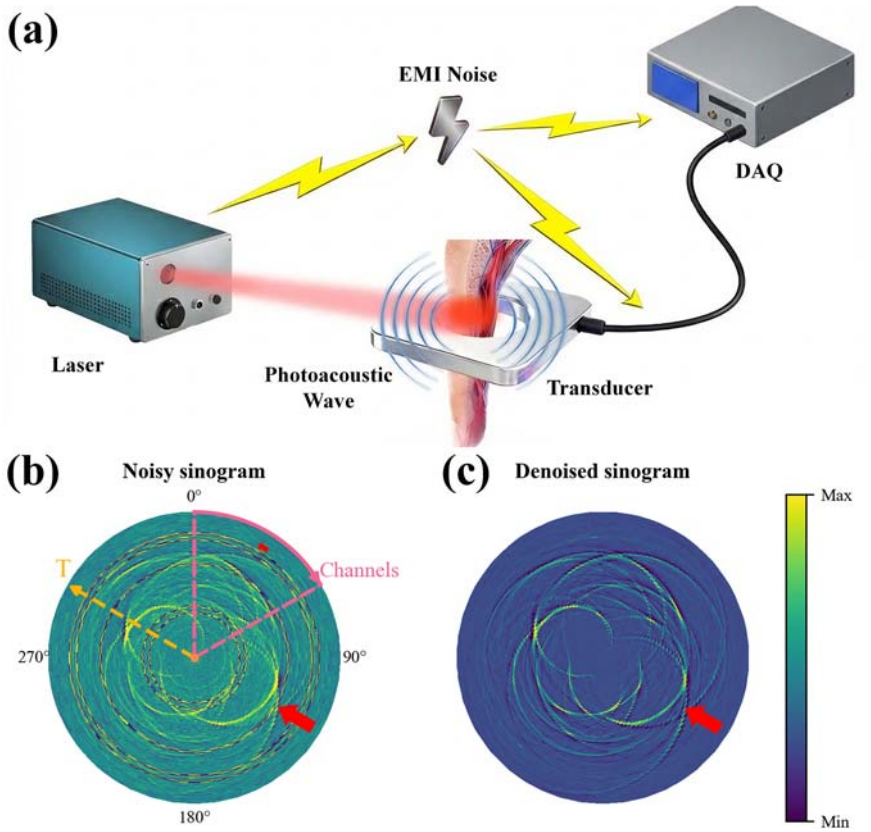


Fig. 6 | Finger PAT system and EMI denoising illustration. a Schematic of the finger PAT system and EMI coupling paths from the laser and ambient environment to the DAQ unit. **b** Noisy sinogram plotted in polar coordinates, where the radial axis encodes the time-of-flight T, and the angular axis encodes the transducer channel index. The orange dashed arrow indicates the time direction, and the pink arc indicates the channel direction. The red arrow highlights a typical stripe-like EMI artifact extending along the channel direction. **c** Denoised sinogram after applying the proposed method. The PA signal trajectories are preserved, while the stripe-like EMI artifacts are suppressed. The grayscale bar on the right shows the normalized amplitude.

detector location $\mathbf{r}_n$ contains the delayed impulse response

$$h'_e\left(t - \frac{\| \mathbf{r}'_m - \mathbf{r}_n \|}{c}\right),$$

where c is the speed of sound and $\| \mathbf{r}'_m - \mathbf{r}_n \|$ is the source-to-detector distance. Thus, even for a single absorber, the arrival time varies with detector position. This acoustic propagation constraint differs from the quasi-common-mode EMI model in Eq. (2).

The proposed SSD framework enhances PA signal fidelity by sequentially addressing noise characteristics in raw channel data. The overall denoising pipeline is illustrated in Fig. 7a. In the first stage, candidate stripe-contaminated time indices are detected from robust cross-channel statistics and corrected by order-statistic replacement. In the second stage, each A-line is processed independently to reduce baseline drift and residual random electronic noise.

Multi-channel filtering for stripe artifact suppression

As illustrated in Fig. 7b, the first-stage procedure consists of two primary steps: threshold selection and stripe artifact removal. Each step is designed to address specific types of noise or distortion present in the raw PA channel data, particularly those introduced by EMI in the acquisition system. The threshold selection module estimates a robust data-driven threshold for identifying candidate structured-noise events. The stripe removal stage targets directional artifacts originating from synchronous EMI across sensor channels and suppresses them by exploiting spatial and temporal priors. Together, these steps effectively reduce EMI-induced stripe artifacts while preserving the spatial coherence of the underlying PA signals.

In this study, "stripe artifact" specifically refers to a near-vertical structure in the channel–time sinogram, where an abrupt transient occurs at approximately the same temporal sample across many receive channels. This definition is different from image-domain striping caused by detector gain mismatch or reconstruction artifacts; here, the stripe is defined in the raw pre-beamforming data domain.

To effectively suppress EMI while limiting attenuation of true PA signals, we introduce a time-domain, quantile-based thresholding strategy that leverages the inherent temporal characteristics of PA acquisition. The core idea lies in exploiting the laser pulse interval during which no true PA signals are present. In this period, large cross-channel fluctuations are treated as candidate interference-dominant events and are used to estimate a robust detection threshold.

Let $X(c, t)$ denote the recorded raw data, where c indexes receive channels and t indexes time samples. We define a pre-signal/background window $\mathcal{T}_0 = \{t < t_0\}$, which is recorded before the expected arrival of tissue-generated PA responses. A one-dimensional cross-channel profile is computed as

$$S_{\text{raw}}(t) = \sum_{c=1}^{N} X(c, t). \tag{4}$$

Large cross-channel deviations in $\mathcal{T}_0$ are treated as candidate EMI-dominant events rather than guaranteed EMI events. To reduce sensitivity to sparse high-amplitude outliers, we use an interquartile-range (IQR) rule to estimate a robust one-sided threshold:

$$\tau = Q_3 + \alpha(Q_3 - Q_1), \tag{5}$$

where Q_1 and Q_3 are the first and third quartiles of $S_{\text{raw}}(t)$ within $\mathcal{T}_0$, and α controls the sensitivity of detection. A smaller α increases sensitivity to weaker EMI events but may increase false-positive masking of PA-dominant samples, whereas a larger α is more conservative but may leave more residual EMI. Therefore, α was treated as a system-dependent operating parameter and should be recalibrated for different PAT systems or EMI environments.

The use of Q_1 and Q_3 follows the standard IQR-based robust outlier rule. Compared with mean- and standard-deviation-based thresholds, the IQR is less affected by sparse high-amplitude EMI spikes in the background window. Because the detection profile is one-sided, with larger values indicating stronger candidate interference, the threshold is applied as an

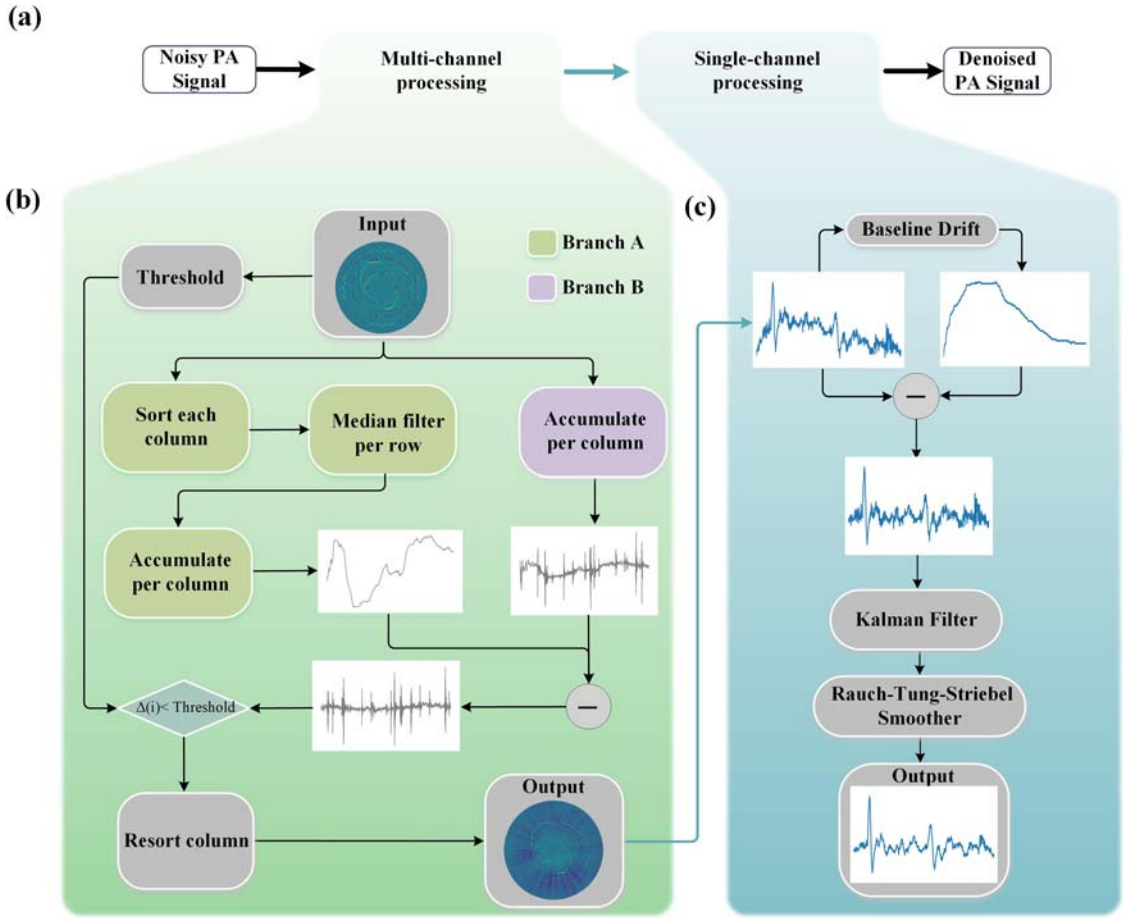


Fig. 7 | Proposed pipeline for EMI removal in PA channel data and its stripe-detection/correction subroutine. a Flow chart of the proposed SSD method for EMI removal with improved PA signal fidelity in the channel data. **b** Detailed procedure for detecting and correcting EMI-induced stripe artifacts. Branches A and B are marked with distinct color-coded blocks. **c** Detailed procedure for single-channel processing.

upper–tail rule:

$$M(t) = \begin{cases} 1, & \text{if } S_{\text{raw}}(t) > \tau, \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

This binary mask flags candidate time indices that are likely dominated by cross-channel EMI and is later used as a prior for stripe correction. The thresholding procedure is illustrated in Fig. 8.

In this stage, we aim to remove stripe-like artifacts in the sinogram caused by temporally synchronized EMI. These artifacts manifest as high-amplitude vertical patterns that span across multiple channels at specific time points, disrupting the spatial coherence expected from genuine PA signals.

We propose a column-sorting correction strategy, which leverages the temporal-spatial characteristics of EMI to identify and suppress such artifacts. The process involves comparing two aggregated signal profiles(i.e., structured denoising and the raw signals) and locating time intervals exhibiting strong discrepancies. The contaminated regions are then corrected by spatially reordered and filtered data.

First, for each time index t, the channel vector $X(:,t)$ is sorted across channels to obtain $X^{\text{sort}}(:,t)$. A one-dimensional median filter with window length L is then applied along the sorted channel dimension:

$$\tilde{X}^{\text{sort}}(:,t) = \text{median_filter}\left(X^{\text{sort}}(:,t), L\right). \tag{7}$$

The original sorting indices are stored so that the filtered vector can later be mapped back to the original channel order.

Next, the raw and filtered cross-channel profiles are computed as

$$S_{\text{raw}}(t) = \sum_{c=1}^{N} X(c,t), \qquad S_{\text{filt}}(t) = \sum_{c=1}^{N} \tilde{X}(c,t), \tag{8}$$

where $\tilde{X}(c,t)$ denotes the reordered version of $\tilde{X}^{\text{sort}}(:,t)$. The deviation profile is then defined as

$$\Delta(t) = |S_{\text{raw}}(t) - S_{\text{filt}}(t)|. \tag{9}$$

A candidate stripe-contaminated time index is detected as

$$M_{\text{stripe}}(t) = \begin{cases} 1, & \Delta(t) > \tau, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

Finally, for each detected time index t, the raw channel vector is replaced by the reordered filtered vector:

$$X_{\text{corr}}(:,t) = \text{resort}\left(\tilde{X}^{\text{sort}}(:,t)\right). \tag{11}$$

For non-detected time indices, $X_{\text{corr}}(:,t) = X(:,t)$. This operation attenuates near-synchronous stripe-like events while preserving the original samples at time indices not flagged by the detector.

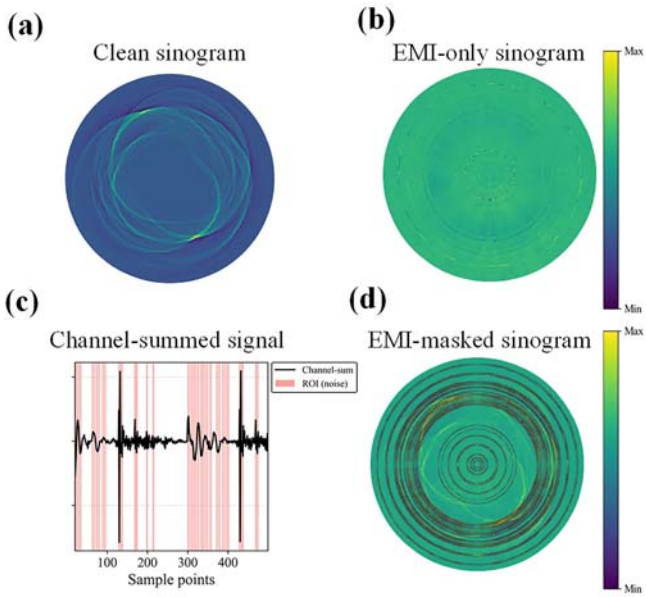


Fig. 8 | Illustration of EMI suppression using quantile-based thresholding. **a** Clean ground-truth sinogram. **b** Simulated EMI-only sinogram representing pre-laser interference. **c** Channel-summed signal with a quantile threshold derived from (**b**). **d** Final sinogram after applying the threshold-based mask to suppress EMI frames; the red strips indicate the detected EMI-contaminated regions.

Single-channel filtering for baseline and gaussian noise removal

Although multi-channel filtering effectively eliminates globally correlated artifacts such as EMI-induced stripes, residual noise components that are localized to individual channels may still persist. These include baseline drift and stochastic noise, which cannot be fully captured or removed by spatial-correlation-based filtering alone. To address this issue, we introduce a single-channel filtering stage that operates independently on each A-line Fig. 7c, enabling precise correction of channel-specific signal distortions.

After stripe correction, residual low-frequency background components may remain as slowly varying row-wise offsets in the channel–time sinogram. Using X_{corr} as the stripe-corrected raw data, we estimate this shared background component across channels using a row-wise median filter:

$$B(:,t) = \mathrm{median_filter}\left(X_{\mathrm{corr}}(:,t), L_b\right). \tag{12}$$

The baseline-corrected data are then obtained as

$$X_{\mathrm{base}}(:,t) = X_{\mathrm{corr}}(:,t) - B(:,t). \tag{13}$$

After stripe correction and baseline removal, the remaining degradation is mainly composed of temporally correlated residual drift and random electronic fluctuations along each A-line. A fixed-window temporal smoother can reduce such fluctuations but may blur short PA transients and distort arrival-time information. A recursive formulation is therefore used to combine the previous state estimate with the current measurement, allowing the correction strength to adapt to local innovations while preserving transient PA waveforms. The recursive stage is not intended to identify cross-channel stripe EMI; that component is handled by the preceding structured stripe-correction stage.

We initialize from the first sample and run the standard prediction-update recursion to output a denoised trace for every A-line; stacking these traces yields the denoised sinogram. Inspired by[5], we introduce a self-adaptive estimation scheme that determines both Q and R directly from the denoised signal, eliminating the need for manual tuning or additional background acquisitions.

The measurement noise variance R is estimated for each A-line independently using the pre-signal region. For each channel j, a sliding window sized n is defined before the arrival of the PA signal:

$$R_j = \frac{1}{n}\sum_{i=1}^{n}\left(x_{i,j} - \bar{x}_j\right)^2, \tag{14}$$

where $x_{i,j}$ denotes the signal at time i for channel j, and $\bar{x}_j$ is the local mean over the first n samples. This region is assumed to contain no PA signal, and thus represents the noise floor.

Unlike the original logarithmic search proposed in[5], we adopt a two-stage adaptive search strategy for Q: (i) a coarse-to-fine dynamic search is performed over a candidate range, (ii) the signal-to-noise quality is evaluated for each candidate via a forward Kalman filter followed by a backward Rauch-Tung-Striebel (RTS) smoother[30]. For each candidate Q, the denoised signal $\hat{x}(t)$ is assessed using a quasi-SNR (QSNR) metric defined as:

$$\mathrm{QSNR} = \frac{\max_t |\hat{x}(t)|}{\max_{t\in\mathcal{T}_0} |\hat{x}(t)|}, \tag{15}$$

The optimal Q_i for each A-line is selected by maximizing QSNR. Then, a global system noise variance Q is obtained by computing the median over all optimal Q_i values:

$$Q = \mathrm{median}\left(\{Q_i\}_{i=1}^{n}\right), \tag{16}$$

The final adaptive values of Q and R are applied to a standard Kalman filter, defined by the following recursive update equations:

$$\begin{aligned}
\hat{x}_{k|k-1} &= \hat{x}_{k-1|k-1},\\
P_{k|k-1} &= P_{k-1|k-1} + Q,\\
K_k &= \frac{P_{k|k-1}}{P_{k|k-1} + R},\\
\hat{x}_{k|k} &= \hat{x}_{k|k-1} + K_k\left(y_k - \hat{x}_{k|k-1}\right),\\
P_{k|k} &= (1 - K_k)P_{k|k-1}.
\end{aligned} \tag{17}$$

This adaptive filtering strategy ensures that the algorithm dynamically adjusts to signal characteristics on a per-channel basis. As a result, it

effectively suppresses residual baseline fluctuations and high-frequency components without compromising the temporal resolution of the PA signal.

Because $Q > 0$ and $R > 0$, the Kalman gain satisfies $0 < K_k < 1$. The update can therefore be written as

$$\hat{x}_{k|k} = (1 - K_k)\hat{x}_{k|k-1} + K_k y_k. \tag{18}$$

Thus, each posterior estimate is a convex combination of the prediction and the current measurement. Consequently, bounded measurements produce bounded state estimates, and the scalar recursion cannot generate unbounded amplitude growth under positive finite covariance estimates. In practice, this boundedness reduces the risk of numerical overflow and unstable amplitude excursions, providing more predictable input for subsequent DAS reconstruction.

Datasets

We used a blood vessel phantom as the initial PA pressure distribution, and the corresponding PA signals were numerically generated using the k-Wave toolbox[31]. We configured a virtual ring transducer array with 128 elements and an 18 mm radius, set the initial pressure map to 26 × 26 mm with 256 × 256 pixels. The center frequency of the array was set as 5 MHz with 80% fractional bandwidth, and the sound velocity of the propagation medium was 1500 m/s. To emulate realistic EMI, we recorded background noise from the acquisition chain with the laser shuttered and no target present, and additively superimposed this measured trace onto the simulated sinograms.

To assess the proposed EMI-suppression method in real-world settings, we acquired PA signals from two in vivo experiments: a human finger experiment and a live zebrafish experiment. Both experiments were conducted in water using a customized 128-channel full-ring PAT array with a ring radius of 30 mm. For illumination, a 720 nm pulsed laser was delivered through a fiber bundle to provide uniform circular illumination around the transducer. PA signals were sampled at 40 MSa/s and processed offline using the same denoising pipeline[32].

The human finger experiment was performed on a healthy adult volunteer. The study involving human participants was approved and monitored by the Ethics Review Board of ShanghaiTech University under approval number SISTIRB2022-002. Written informed consent to participate in the study was obtained from the volunteer before the experiment. All experimental procedures were reviewed to ensure that no safety concerns were identified and were conducted in accordance with the Declaration of Helsinki.

Zebrafish were maintained under standard laboratory conditions at the ShanghaiTech University animal facility. Adult fish were kept in a recirculating aquaculture system under a standard light/dark cycle. All zebrafish experiments were approved by the Institutional Animal Care and Use Committee (IACUC) of ShanghaiTech University under protocol number 20200323002, and were performed in accordance with relevant institutional guidelines and regulations. Photoacoustic imaging was performed as a non-invasive imaging procedure under the approved animal protocol. No chemical anesthesia was used during imaging. After imaging, zebrafish were returned to the animal facility without euthanasia, in accordance with the approved protocol. Randomization and blinding were not applied because the zebrafish experiments were used for imaging-data acquisition and algorithm validation rather than treatment-group comparison.

For the human finger and zebrafish measurements, EMI was not artificially injected. Its occurrence was verified from the acquired raw channel data before denoising. The interference appeared as abrupt high-amplitude events shared by multiple channels at approximately the same sample indices, forming near-vertical stripe-like structures in the channel time sinograms. These events did not follow the channel-dependent time-of-arrival pattern expected from acoustic PA propagation, and their cross-channel synchrony was further quantified in "Quantitative validation of inter-channel EMI synchrony".

Comparison methods

To evaluate the effectiveness of SSD for suppressing EMI-related artifacts in PA raw channel data, we compared it with five representative denoising baselines: Block-Matching and 3D Filtering (BM3D)[33], one-dimensional guided filtering (1D–GF; labelled DF in figure panels)[34], weighted least squares filtering (WLSF)[35], non-local means (NLM), and normalized least-mean-square adaptive noise cancellation (NLMS-ANC).

BM3D was included as a classical non-local image-denoising baseline. It suppresses local random fluctuations by collaborative filtering of similar patches, but it does not explicitly model quasi-synchronous cross-channel EMI. The 1D-GF method was included as a guided-filtering-based stripe-removal baseline. It estimates stripe components through directional smoothing and local linear filtering. WLSF was included as a weighted least-squares stripe-removal baseline, where edge-preserving smoothing and weighted regression are used to suppress structured stripe components.

NLM was included as a representative non-local denoising baseline. It performs patch-similarity-based averaging and can effectively reduce local random fluctuations, but it does not explicitly exploit the near-synchronous cross-channel structure of EMI. NLMS-ANC was included as a representative adaptive noise-cancellation method for suppressing strong interference. Because the present PAT system does not provide an independent hardware reference noise channel, the reference signal was estimated from background/noise-dominant raw measurements. This implementation represents a practical reference-estimation strategy under the current acquisition configuration, while also highlighting the dependence of adaptive noise cancellation on the availability and quality of the reference signal.

All comparison methods were applied to the same raw-channel inputs and evaluated using identical DAS reconstruction parameters, ROI masks, and quantitative metrics. For methods that operate naturally in the image or two-dimensional sinogram domain, the processed outputs were reconstructed with the same DAS pipeline to ensure a fair comparison.

Additional no-clean-label and self-supervised baseline experiments, including training details and representative outputs, are provided in the Supplementary Information.

Evaluation metrics

For simulated datasets where the clean ground truth is known, we adopt two widely used image quality metrics: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM).

PSNR measures the fidelity of the reconstructed image by quantifying the discrepancy between the denoised image I and the reference I_{ref}. Let MAX_I denote the peak possible pixel value. Then

$$\begin{aligned} \mathrm{PSNR} &= 10\log_{10}\left(\frac{\mathrm{MAX}_I^2}{\mathrm{MSE}}\right), \\ \mathrm{MSE} &= \frac{1}{N}\sum_{p\in\Omega}\left(I(p) - I_{\mathrm{ref}}(p)\right)^2, \end{aligned} \tag{19}$$

where Ω is the set of pixel locations with $|\Omega| = N$. PSNR is reported in decibels (dB).

SSIM evaluates perceived image quality by comparing luminance, contrast, and structural similarity between the denoised and reference images. It is computed as:

$$\mathrm{SSIM}(I, I_{\mathrm{ref}}) = \frac{\left(2\mu_I\mu_{I_{\mathrm{ref}}} + C_1\right)\left(2\sigma_{II_{\mathrm{ref}}} + C_2\right)}{\left(\mu_I^2 + \mu_{I_{\mathrm{ref}}}^2 + C_1\right)\left(\sigma_I^2 + \sigma_{I_{\mathrm{ref}}}^2 + C_2\right)} \tag{20}$$

where μ, σ^2, and $\sigma_{II_{\mathrm{ref}}}$ denote local means, variances, and cross-covariances, respectively; C_1 and C_2 are constants to avoid instability, typically set to $C_1 = (0.01)^2 = 1 \times 10^{-4}$ and $C_2 = (0.03)^2 = 9 \times 10^{-4}$ for images normalized to the range [0, 1].

For real experimental data where the clean ground truth is unavailable, we adopt two reference-free metrics to quantify signal quality: Generalized

Contrast-to-Noise Ratio (gCNR) and A-line SNR. For the real finger and zebrafish datasets, image-domain signal and background ROIs were independently defined by a blinded evaluator using only the raw/noisy DAS images for gCNR computation. The evaluator was not involved in SSD development or parameter tuning and did not have access to SSD-denoised images, method labels, or quantitative results during ROI selection. The resulting ROI masks were fixed and applied unchanged to all methods. A-line SNR was computed from selected one-dimensional signal and background/noise windows on the corresponding A-lines, with the same A-line window definitions applied to all compared methods within each frame. For multi-frame in vivo analysis, gCNR and A-line SNR were computed frame by frame and reported as mean ± standard deviation across frames.

To support these frame-wise summary results, extended frame-wise statistics and ROI annotations are provided in the Supplementary Information.

For the simulation dataset, we additionally quantified potential over-suppression introduced by the first-stage stripe-detection/correction module. Since the stripe mask is defined at the time-index level, the false-positive analysis is also performed at the same level.

Let

$$\Delta(t) = |S_{\mathrm{raw}}(t) - S_{\mathrm{filt}}(t)|$$

denote the stripe-detection profile. Using the simulation ground truth, we define

$$E_{\mathrm{sig}}(t) = \sum_{c} S_{\mathrm{clean}}(c,t)^2, \qquad E_{\mathrm{emi}}(t) = \sum_{c} S_{\mathrm{emi}}(c,t)^2.$$

A time index is labeled as signal-dominant when $E_{\mathrm{sig}}(t) > E_{\mathrm{emi}}(t)$. The false-positive rate is defined as

$$\mathrm{FPR} = P\Big(M_{\mathrm{stripe}}(t) = 1 | E_{\mathrm{sig}}(t) > E_{\mathrm{emi}}(t)\Big).$$

To quantify signal attenuation caused by stripe correction, we further compute

$$\eta_{\mathrm{loss}} = 1 - \frac{\left\| S_{\mathrm{clean}}^{\mathrm{corr}} \right\|_2^2}{\left\| S_{\mathrm{clean}} \right\|_2^2}.$$

gCNR assesses the separability between signal and background distributions[36]. Let $p_s(x)$ and $p_b(x)$ denote the probability density functions of the signal and background regions, respectively. The gCNR is defined as:

$$\mathrm{gCNR} = 1 - \int \min\{p_s(x), p_b(x)\} dx \tag{21}$$

This definition quantifies the non-overlapping area between the two distributions. A higher gCNR value indicates better contrast and improved distinguishability between signal and background.

A-line SNR measures the energy ratio between the selected A-line signal window and the surrounding background/noise window. It is calculated as:

$$\mathrm{SNR} = 10\log_{10}\left(\frac{\mu_s^2}{\sigma_b^2}\right) \tag{22}$$

where μ_s is the mean intensity in the selected one-dimensional signal window and σ_b^2 is the variance in the selected one-dimensional background/noise window on the corresponding A-line. A higher SNR implies better signal clarity.

Data availability

The datasets generated and analyzed during the current study are not publicly available due to institutional data-management restrictions and privacy considerations related to in vivo experimental data, but are available from the corresponding author on reasonable request.

Code availability

The code used in this study is not publicly available due to ongoing project development and institutional data-management restrictions, but is available from the corresponding author on reasonable request.

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Author contributions

R.S. conceived the study, developed the SSD algorithm, performed simulations and experiments, and drafted the manuscript. F.Z. contributed to system design, data acquisition, and result analysis. Z.H. assisted with algorithm implementation and data processing. Y.X. contributed to experimental setup and data acquisition. S.L. contributed to system integration and hardware debugging. Y.Z. assisted with data analysis and figure preparation. Y.J. contributed to interpretation of results and manuscript revision. S.T. provided support on biological experiments and zebrafish handling. H.L. co-supervised the project and contributed to the interpretation of the results and manuscript revision. X.C. co-supervised the project and contributed to the interpretation of the results and manuscript revision. F.G. supervised the overall project, provided critical guidance, and revised the manuscript.All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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